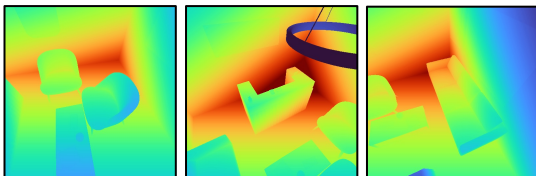
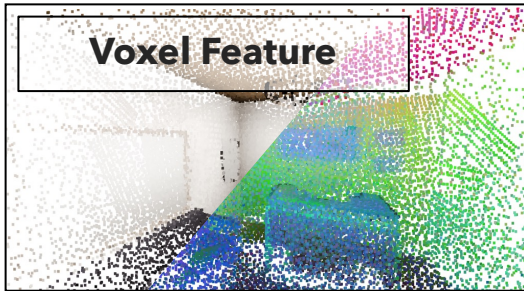


## Input Images

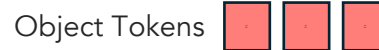
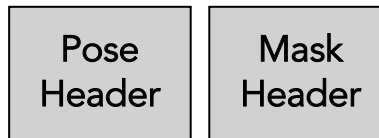
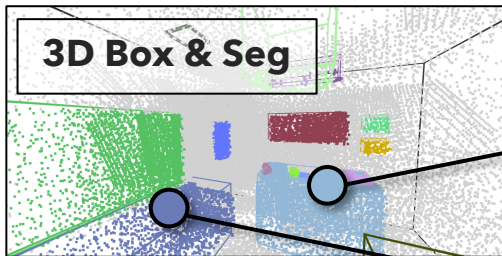


## Voxel Feature



3D Lifting & Feature Extraction

## 3D Box & Seg



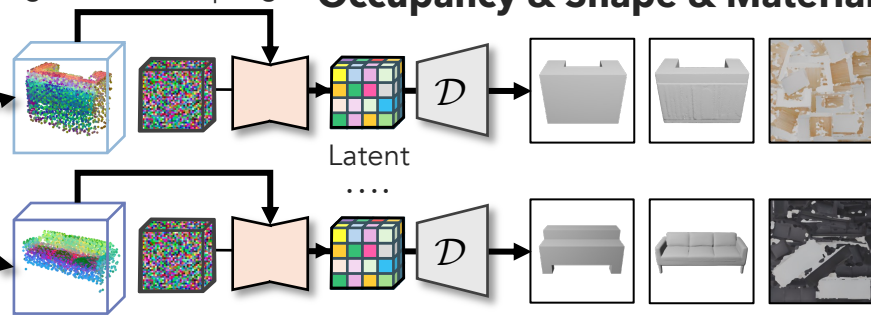
## Transformer Backbone



Voxel Tokens

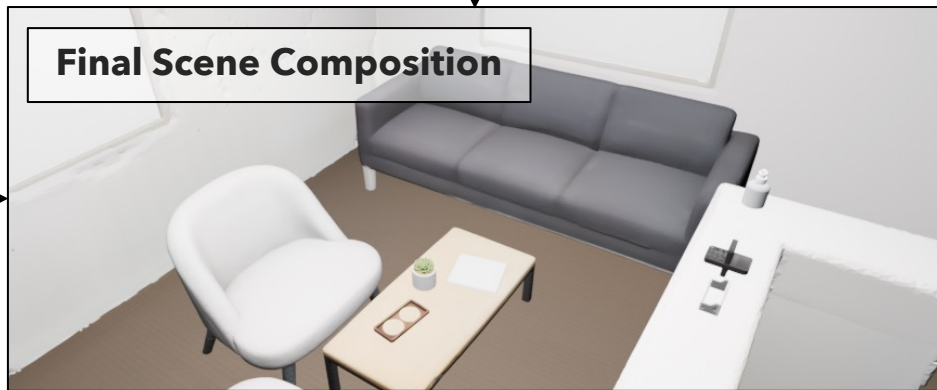
FIRE3D-Perception Network

## Single Batch Sampling



## Occupancy & Shape & Material

## Final Scene Composition



FIRE3D-Shape Network